Lesson 7

If Statements: Choosing a Path



- In real life, we make choices like "If it's raining, take an umbrella."
- In Python, we do the same thing using If Statements.
- If a condition is True, the code runs.
- If it is False, the computer skips it!

The Guard

An 'If' statement is like a guard. It asks a True/False question using operators. It only lets the instructions through if the answer is a solid "Yes" (True).



How to Write an If Statement

- Start with the keyword 

- Add your condition (the comparison you want to check).
- End the line with a colon
  Do not forget the colon! 

- Indent the code you want to run!

Indentation Rules

4 Spaces

Usually 4 spaces or 1 tab. It shows Python which lines belong to the 'if'.

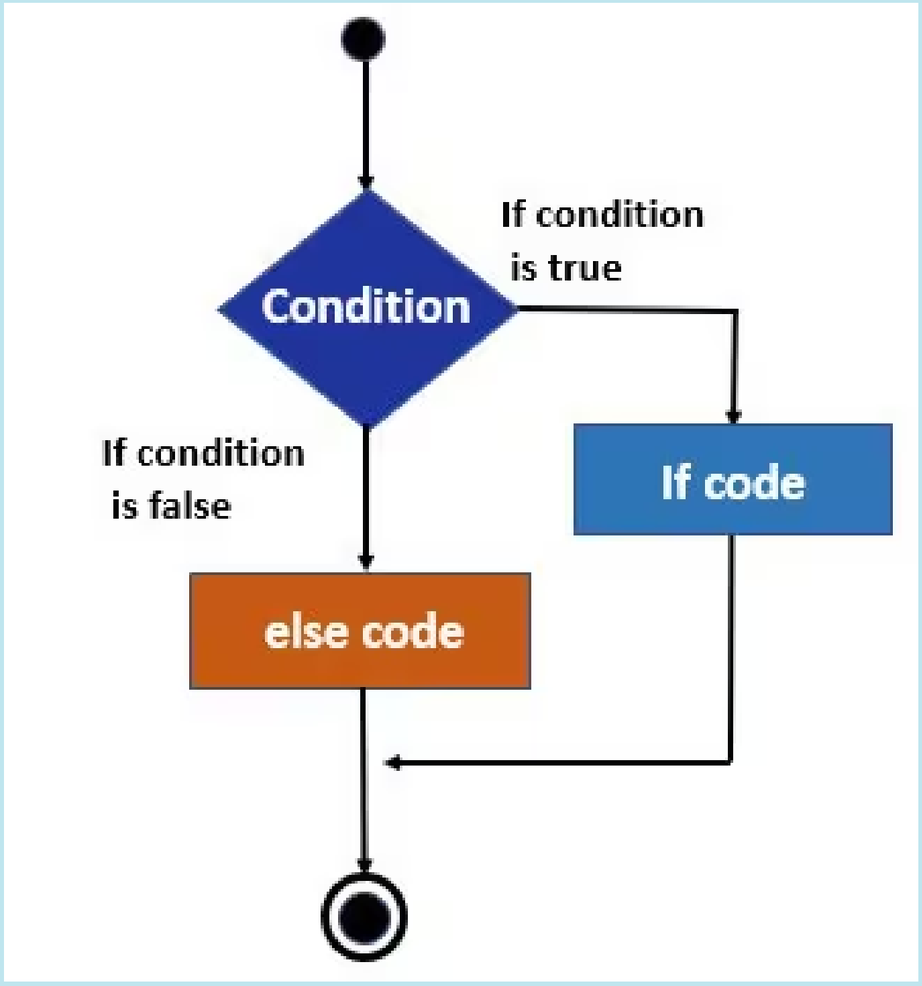
Inside the Block

Every indented line under the 'if' will run only if the condition is True.

No Indent = Error

If you don't indent, Python won't know those lines are part of the decision!

The Flow of Code



Game Logic

Unlocking Levels

Imagine you need 100 coins to unlock a door:

```
if coins >= 100:
  print("Door Unlocked!")
```

If you have 99 coins, nothing happens. The math must be right!

Common Mistakes

Missing Colon

Forgetting the `:` is the #1 mistake beginners make!

Bad Indent

If the code isn't pushed in, Python won't run it as part of the 'if'.

Single Equal

Remember: Use `==` to check, not `=` to set!

If - Else: Game Logic



The "Otherwise" Plan

An `else` statement is your Plan B. It tells Python what to do when the `if` condition is **False**.

It's like saying: "If it's sunny, wear a hat. **Otherwise (Else)**, take an umbrella!"

Pass/Fail Test



```
score = 75

if score >= 50:
  print("You Passed!")
else:
  print("Try Again!")
```

Since 75 is greater than 50, only the first message prints. The "Else" part is ignored!

Activity the Cinema Bot

The Challenge

Create a program that asks for a user's age. If they are 13 or older, tell them to "Buy a Ticket." Otherwise, tell them to "Bring a Parent!"



Hints

1. Use `int(input())` for age.
2. Don't forget the colon after else!
3. Indent your print statements.

Cinema Bot Answer

```
# Ask for the user's age and convert it to an integer
age = int(input("What is your age? "))

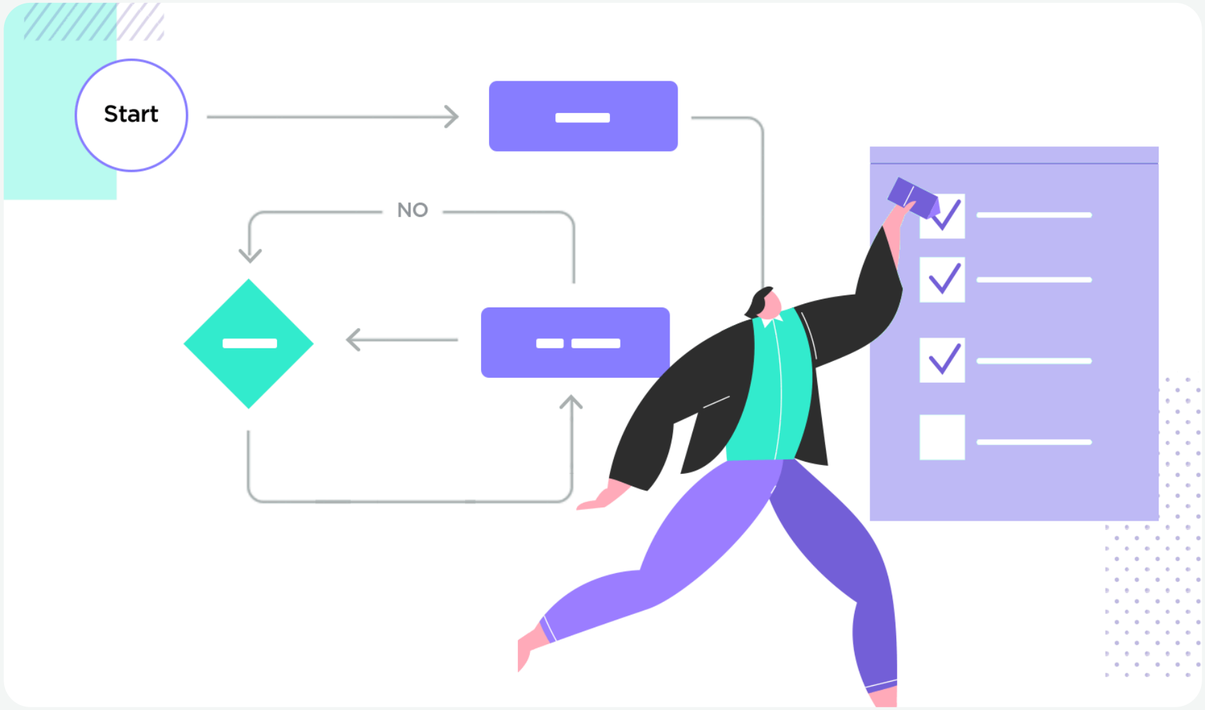
# Check if they are 13 or older
if age >= 13:
    print("Buy a Ticket.")
else:
    print("Bring a Parent!")
```

The One-Way Rule

Either / Or

In an `if-else` block, exactly **ONE** path will always be taken. Python will never run both, and it will never skip both.

Think of it like a light switch: It is either ON or OFF. There is no middle ground!



The Elif Statement

When two choices aren't enough, `elif` (short for else-if) lets you add as many extra conditions as you want!

You can link an **infinite** number of `elif` blocks together between your initial `if` and your final fallback `else`.

```
score = 85

if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
else:
    print("Grade: C")
```

If vs Elif



What is the difference between if and elif?

two separate ifs: both run
If we have elif and if, only one runs, even if both conditions are true

If vs Elif



Logical Operator logic

When you chain multiple if statements, Python treats them as entirely separate questions. It will check every single one of them

When you use elif, the conditions are linked together. Python looks for the first true condition, runs that block, and immediately skips everything else

And vs Or

What are Nested Conditions?

Decision Layers

A **Nested Condition** is an `if` statement inside another `if` statement.

Think of it like a security checkpoint. To get the prize, you have to unlock the front door first, and then open a secret safe inside!

Nested if

Nested Conditions: Layered logic (If inside If). The inner check is only reached if the outer check passes. Watch your double indentation!

Nested if statements vs multiple independent ifs

Use case: When a secondary check is completely pointless unless the primary check is satisfied

ex. You need to pay and have permission to go on field trip. If the first condition of having permission is not satisifed, there is no point in checking if you paid.

Example: Game

Tracking the Shifts

Notice the extra space on line 5!

Because the second `if` is tucked inside the first one, it gets shifted inward twice (8 spaces total for its action!).

```
has_membership = True
level = 15

if has_membership == True:
    if level >= 10:
        print("Ranked mode unlocked!")
    else:
        print("Reach level 10 first.")
```

Program

You need to pay and have
permission to go on a field trip.

Ask user if they have permission
and if they paid.

Print if they can go or not

Nested Conditions Recap

Boxes inside Boxes

Nested logic means putting checks inside other checks to filter specific rules.

Double Indent

The inner block must step right **again** to show it belongs to the second layer.

Keep it Clean

Avoid building a giant pyramid. Limit your nesting layers so your script stays scannable.

Driving eligibility

Ask user if they have license and if they are older than 18, print their driving eligibility. Both conditions should be met to drive.

Numbers

Ask user to enter a number. Check if it is pos/ neg and even/odd. Print.

CHALLENGE - WE WILL DISCUSS THIS ONE NEXT CLASS

Write a program that converts celcius to farenheit or farenheit to celcius. User should input what they are converting from and converting to and enter the temperature. Print the converted value.

Practice Programs to do at home

Program

Scholarship?

To get a scholarship, the student should have either a 4.0 GPA or more than 29 credits and have atleast 2 extracuricular activites.